Virtual, Augmented And Spatial Computing - Module Descriptor

Bachelor of Science (Honours) in Creative Digital Computing

2026-08-28

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Virtual, Augmented And Spatial Computing

Short title	VR, AR & Spatial Computing
Slot	Stage 3, Semester 1
Credits	5 ECTS
Type	Mandatory
NFQ level	7
Author	John Jennings
Stream	Game Development
Status	Linked to authored course

Virtual and augmented reality, also known as immersive reality, is any technology that replicates an environment, real or imagined, and simulates a user's physical presence and environment to allow for user interaction. Spatial computing enables immersive interaction by understanding the physical world, knowing and communicating our relation to places in that world, and navigating through those places..

Learning outcomes

Code	Outcome
MLO1	Demonstrate a critical understanding of the concepts and principles of 3D systems and augmented and virtual reality in the context of spatial computing
MLO2	Explore, critique and deconstruct modern VR, AR, and spatial computing experiences.
MLO3	Demonstrate theoretical understanding of, and practical competence in the use of advanced input and output devices & sensors in an immersive environment.
MLO4	Apply appropriate principles of human-computer interaction to the construction and evaluation of virtual and augmented reality and spatial computing systems.

Indicative syllabus

- VR & AR Systems
  - Hardware / Software
  - Headsets
  - Optical systems
  - Tracking systems
  - Biometrics
- Perception & Presence
  - Immersion & Presence
  - Reality Trade-offs
  - Health Effects & Mitigation
  - Perception Models and Processes
  - Distal & Promixal Stimuli
  - Perceptual Modalities
- Design Guidelines
  - Interaction
  - XR Interaction
  - Motion & Gesture Tracking

  - Interaction Patterns & Techniques

- Content Creation
  - Environment Design
  - Affecting Behaviour
  - Content Design Guidelines

Assessment

Continuous Assessment **100%** · Final Exam **0%**

Component	Weight	Type	Outcomes assessed
Practical Evaluation CA Lab Work	30%	CA	MLO MLO1, MLO MLO2, MLO MLO3
Case Study	30%	CA	MLO MLO3, MLO MLO4
Create an interactive immersive VR/AR application	40%	CA	MLO MLO1, MLO MLO2, MLO MLO3, MLO MLO4

Recommended reading

- Cronin, Irena and Scoble, Robert and Wozniak, Steve. *The Infinite Retina* (2020). Packt Publishing
- Dalton, Jeremy. *Reality Check: How Immersive Technologies Can Transform Your Business* (2021). KoganPage
- Duckham, Matt. *Decentralized Spatial Computing: Foundations of Geosensor Networks* (2013). Springer
- Hillmann, Cornel. *UX for XR: User Experience Design and Strategies for Immersive Technologies* (2021). Apress
- LaViola, Joseph J. and Kruijff, Ernst and McMahan, Ryan P. and Bowman, Doug A. and Poupyrev, Ivan. *3D User Interfaces: Theory and Practice* (2017). Addison-Wesley
- Linowes, Jonathan. *Unity 2020 Virtual Reality Projects: Learn VR Development by Building Immersive Applications and Games with Unity 2019.4 and Later Versions* (2020). Packt
- *Creating Augmented and Virtual Realities: Theory and Practice for next-Generation Spatial Computing* (2019). O'Reilly Media
- RUBIN, PETER. *FUTURE PRESENCE: How Virtual Reality Is Changing Human Connection, Intimacy, and the Limits of... Ordinary Life* (2020). HARPERONE
- Schmalstieg, Dieter and H"ollerer, Tobias. *Augmented Reality: Principles and Practice* (2016). Addison-Wesley

- Shekhar, Shashi and Vold, Pamela. *Spatial Computing* (2019). The MIT Press
- Stevens, Ren'ee. *Designing Immersive 3D Experiences: A Designers Guide to Creating Realistic 3D Experiences for Extended Reality* (2021). New Riders

Programme outcome coverage

Module LO	Programme outcome	Level
MLO1	PLO1	indirect
MLO2	PLO1	indirect
MLO3	PLO1	indirect
MLO4	PLO1	indirect
MLO1	PLO2	indirect
MLO2	PLO2	indirect
MLO3	PLO2	indirect
MLO4	PLO2	indirect
MLO1	PLO3	indirect
MLO2	PLO3	indirect
MLO3	PLO3	indirect
MLO4	PLO3	indirect
MLO1	PLO4	indirect
MLO2	PLO4	indirect
MLO3	PLO4	indirect
MLO4	PLO4	indirect
MLO1	PLO5	indirect
MLO2	PLO5	indirect
MLO3	PLO5	indirect
MLO4	PLO5	indirect
MLO1	PLO7	indirect
MLO2	PLO7	indirect
MLO3	PLO7	indirect
MLO4	PLO7	indirect
MLO1	PLO8	indirect
MLO2	PLO8	indirect
MLO3	PLO8	indirect
MLO4	PLO8	indirect